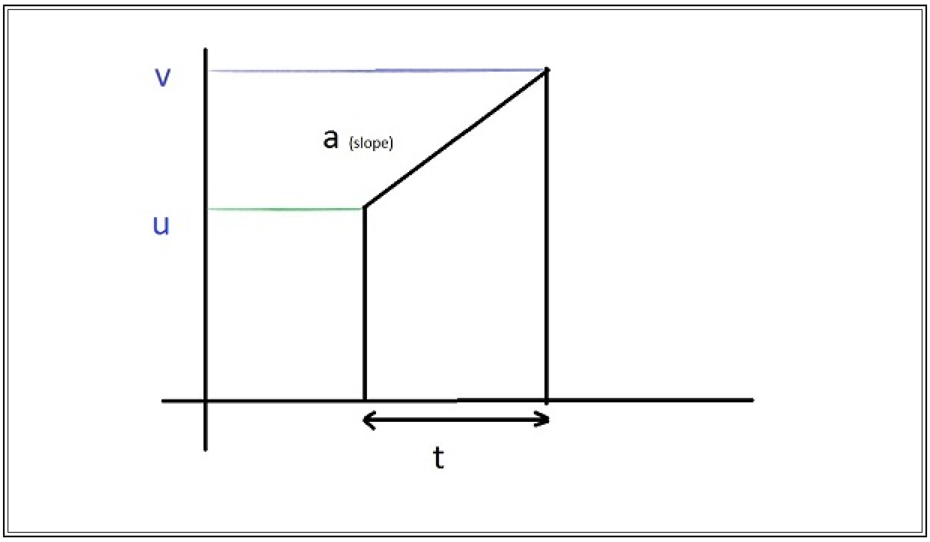
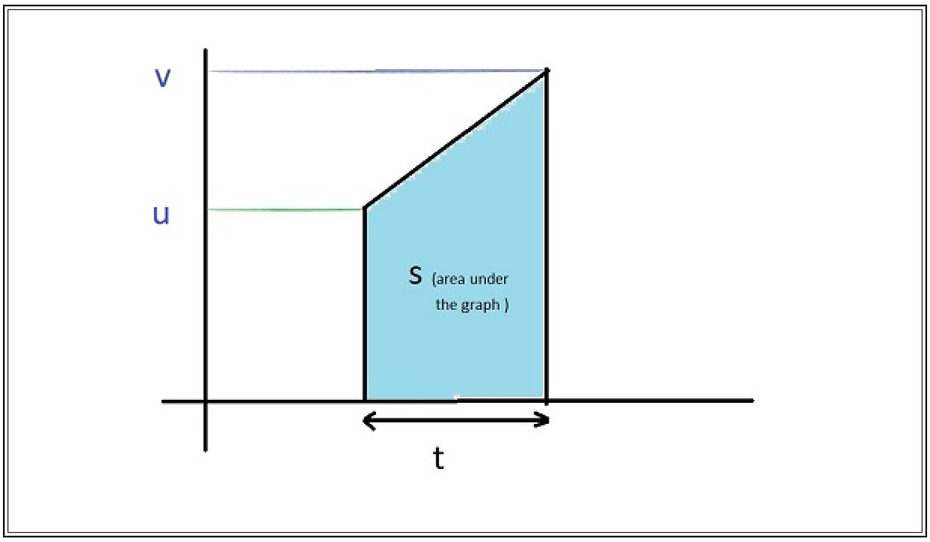
1. Derivation from v-t graph { $v = u + at$ }



We define the slope of v-t graph as a, so it gives $a = \dfrac{v-u}{t}$

Manipulating the form of this equation, we get $v = u + at$, which is newton's first equation of motion in scalar form.

2. Derivation from v-t graph { $s = ut + \dfrac{1}{2}at^2$ }



Area under the v-t graph is dispacement(s)

Area of a trapezium is sum of parallel sides X distance between them

$\Rightarrow s = \dfrac{v+u}{2}t$

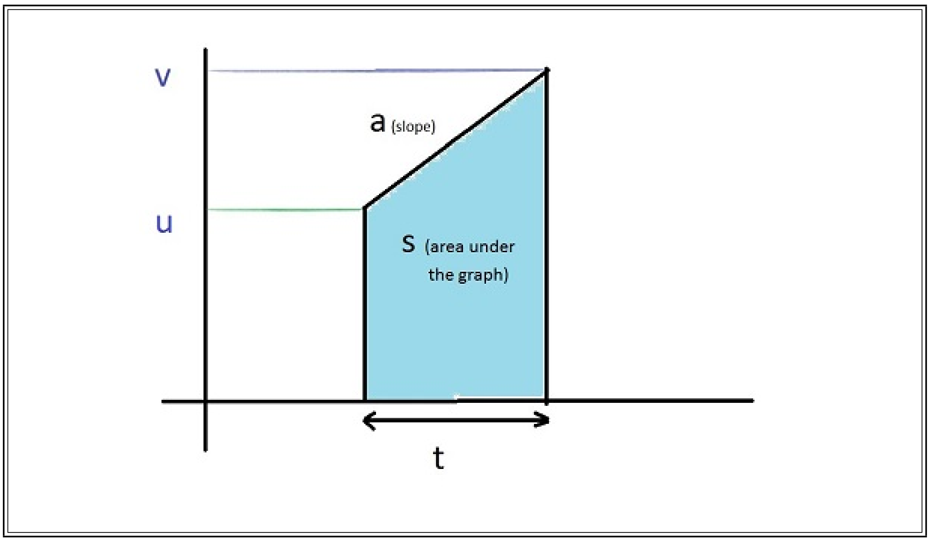
Substituting, the previously derived newton's first equation of motion in scalar form

$\Rightarrow s = \dfrac{u+at+u}{2}t$

OR

$s = ut + \dfrac{1}{2}at^2$

3. Derivation from v-t graph { $v^2 - u^2 = 2as$ }



In deriving the newton's third equation by graph, two values are taken from the graph

$a = \frac{v-u}{t}$ and $s = \frac{v+u}{2}t$

Multiplying both the equations,

$as = \frac{v^2 - u^2}{2}$

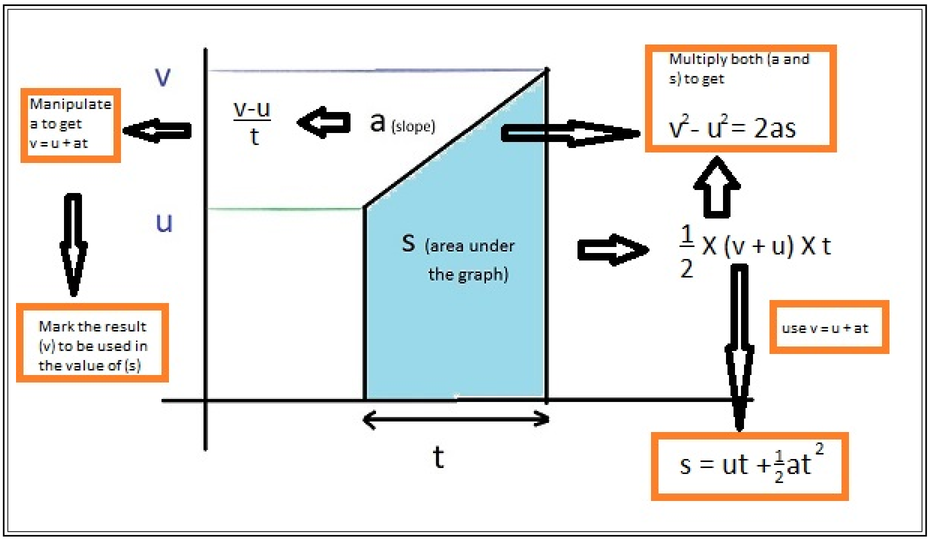
OR

$v^2 - u^2 = 2as$

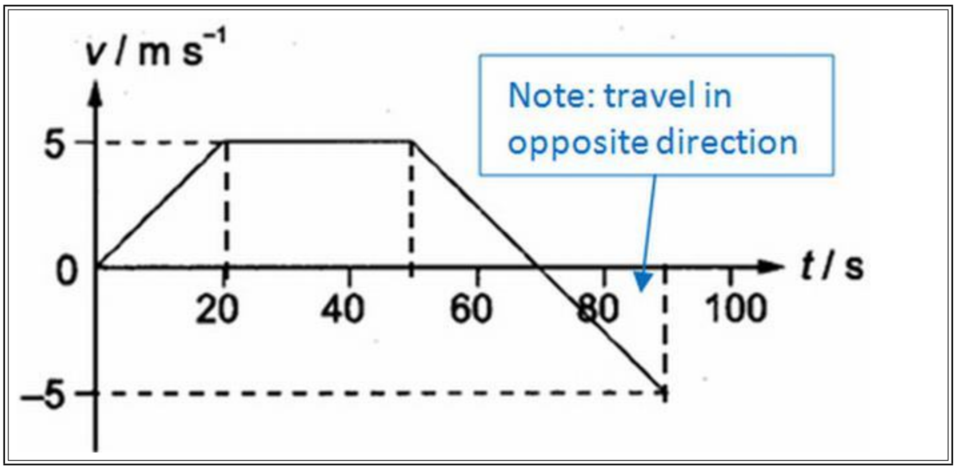
4. Result from Area Under the v-t graph, $s = v_{av}t = \frac{v+u}{2}t$

5. Manipulating $v = u+at$ as $u = v-at$ and substituting in $s = \frac{v+u}{2}t$ gives $s = \frac{v-at+v}{2}t \Rightarrow s = vt - \frac{1}{2}at^2$

The Complete Derivation



Example : A girl starts from rest and travels along a straight line. The diagram below shows the velocity-time graph of the girl from 0 s to 90 s.



(a) Describe the motion of the girl from 0 s to 40 s.

(b) Find the average velocity of the girl in the first 70 s.

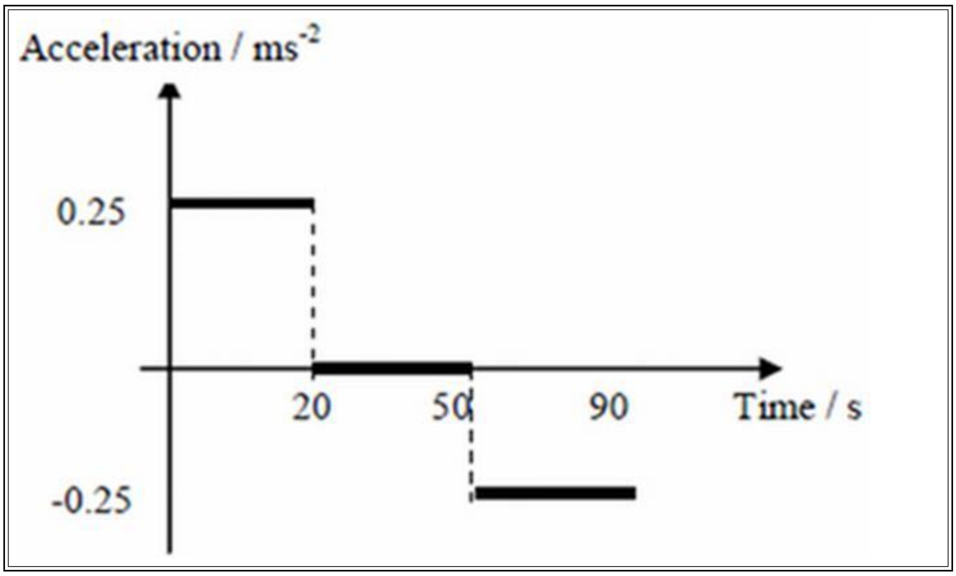
(c) Draw the acceleration-time graph of the girl from 0 s to 90 s.

(d) Find the displacement of the girl from the starting point to the position at 90 s.

Solution : (a) From 0 s to 20 s, she travels with constant acceleration of 0.25 ms^(-2); From 20 s to 40 s, she travels at constant velocity of 5 m/s.

(b) Total displacement = area under v-t graph = $\frac{1}{2}$ (30 + 70) * 5 = 250 m Average velocity = 250/70 = 3.57 m/s

(c) Acceleration is the gradient of v-t graph.

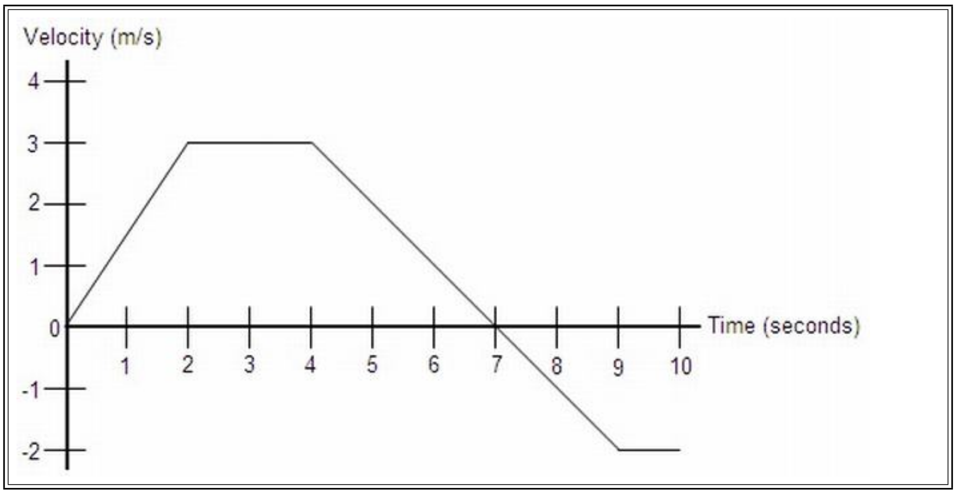


(d) Displacement = area under v-t graph = 250 – $\frac{1}{2}$ * 20 * 5 = 200 m

Tips: (1) It is very easy to mix up kinematics graphs. The only way to differentiate these graphs is to look at the y-axis!

(2) When describing the motion, do it region by region! And always talk about acceleration or speed / velocity with values if applicable. Refer to example above.

Example : An object's velocity during a 10 second time interval is shown by the graph below:



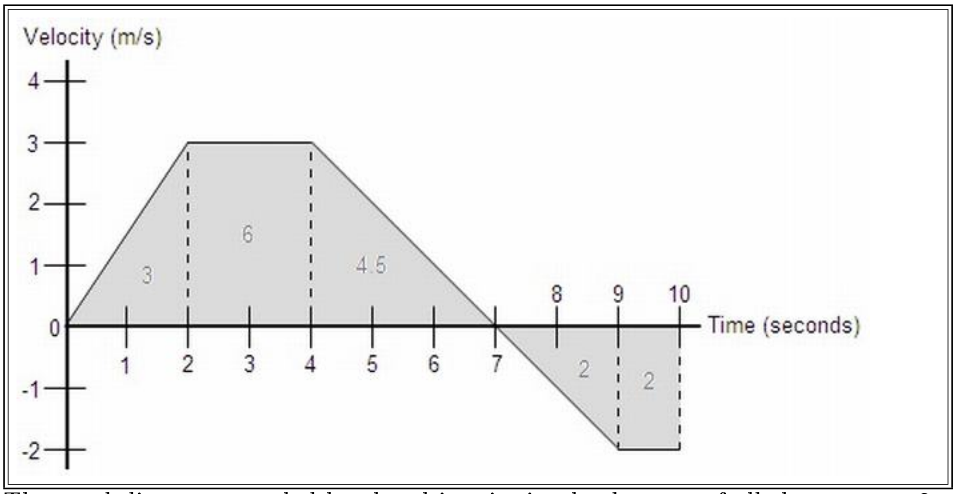
a.) Determine the object's total distance traveled and displacement.

b.) At t = 0, the object's position is x = 2 m. Find the object's position at t = 2, t = 4, t = 7, and t = 10.

c.) What is the object's acceleration at the following times: t = 1, t = 3, and t = 6.

d.) Sketch the corresponding acceleration vs. time graph from t = 0 to t = 10.

Solution : a.) Recall that the equation for velocity is v = x/t. If we solve this for x, we get x = vt. Notice that this is the same as the area of a rectangles whose sides are lengths v and t, so we can determine that the displacement is the area enclosed by the velocity vs. time graph. So, we will find the area of each section under the graph:



The total distance traveled by the object is simply the sum of all these areas: 3 + 6 + 4.5 + 2 + 2 = 17.5 m The displacement is found in a similar fashion, except areas below the x-axis are considered negative: 3 + 6 + 4.5 - 2 - 2 = 9.5 m Interestingly enough, the area enclosed by any function can be represented by a definite integral. For example, if this graph were defined as a function v(t), then the displacement would be the integral from 0 to 10 of v(t)dt, and the total distance traveled would be the integral from 0 to 10 of |v(t)|dt Go to calculus notes

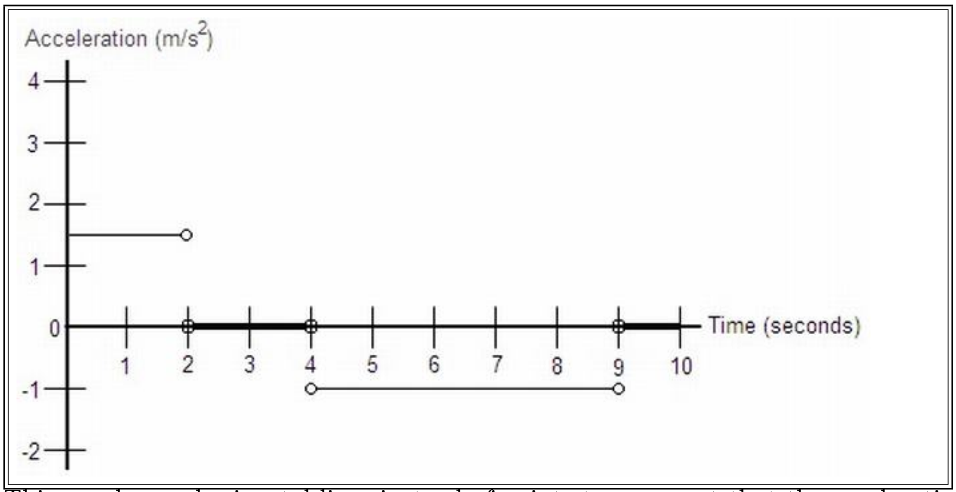
b.) The position of the object at a given point in time can be found in much the same way we found the displacement in part a, except this time we must also add in the initial value given. So: x(2) = 2 + 3 = 5 m x(4) = 2 + 3 + 6 = 11 m x(7) = 2 + 3 + 6 + 4.5 = 15.5 m x(10) = 2 + 3 + 6 + 4.5 - 2 - 2 = 11.5 m

Notice that this can also be done by adding the integral from 0 to t of v(t)dt to the initial value of 2. Go to calculus notes

c.) Like velocity in part b of problem 22, the instantaneous acceleration at any point along one of the graph's line segments is the same as the average acceleration across that line segment. The formula for acceleration is aavg = Δv/Δt = (vf - vi)/(tf - ti), so: a(t) = (vf - vi)/(tf - ti) a(1) = (3 - 0)/(2 - 0) = 3/2 = 1.5 m/s2 a(3) = (3 - 3)/(4 - 2) = 0/2 = 0 m/s2 a(6) = (-2 - 3)/(9 - 4) = -5/5 = -1 m/s2

Similarly to the relationship between velocity and position, the formula for acceleration is the same as the slope formula for a velocity vs. time graph. So, we can say that the slope of any velocity vs. time graph is its acceleration. Notice that this definition defines acceleration as the derivative of velocity. So, it is true that for any velocity function v(t), its derivative is an acceleration function a(t). Also, integration can be used to go from an acceleration function to a position function. Go to calculus notes

d.) We know that the acceleration along each line segment of this velocity vs. time graph is equal to the slope of the line segment. We determined these slopes in part c, so the acceleration graph would look like:



This graph uses horizontal lines instead of points to represent that the acceleration is defined at that value at any point along that section. The open circles at the end of each line segment simply indicate that at those time values, acceleration is not defined at either value represented by the horizontal lines. At these points, acceleration is undefined because it changes instantaneously from one value to the next, which cannot be represented numerically.

Example : On a displacement-time graph, two straight lines make angles of 30° and 60° with the time-axis. The ratio of the velocities represented by them is

a) $1 : \sqrt{3}$

b) 1:3

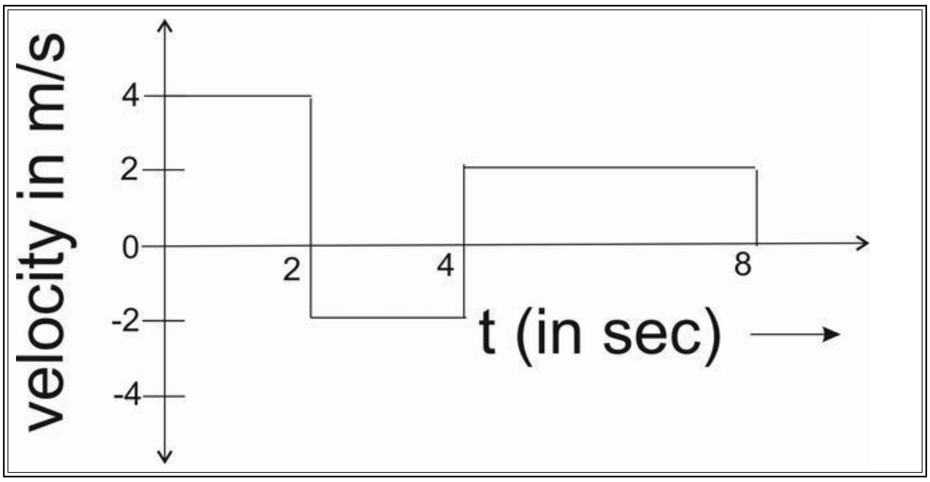
c) $\sqrt{3} : 1$

d) 3:1

{ Hint: The velocity in a displacement-time is given by the slope of the curve. Slope = tan(gent) of angle of inclination of s-t graph. This gives the respective ratios tan 30^o / tan 60^o

Answer: b) is the correct answer. }

Example : A body is moving in a straight line as shown in velocity-time graph. The displacement and distance travelled by body in 8 second are respectively:



a) 12 m, 20 m

b) 20 m, 12 m

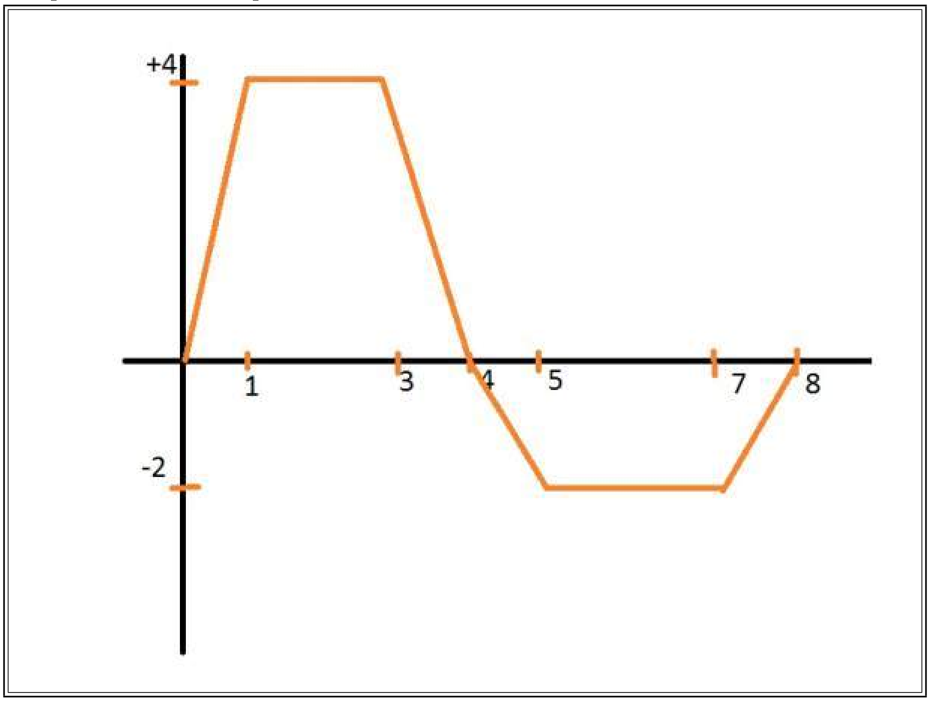
c) 12 m, 12 m

d) 20 m, 20 m

{ Hint: The displacement in a velocity-time graph is given by the area under the graph with proper signs. From 0s - to 2s , the area is 8m . From 2s - to 4s , the area is -4m . From 4s - to 8s , the area is 8m. Adding these 3 values , we get 8m + (-4m) + 8m = 12m. } The distance in a v-t graph is given by the absolute area under the graph. So, taking the absolute values of individual area divisions, we get 8m + 4m + 8m =20m

Answer: a) is the correct answer. }

Example : The velocity-time graph of a particle in linear motion is as shown. Both v and t are in SI units. The displacement of the particle is



a) 6 m

b) 8 m

c) 16 m

d) 18 m

{ Hint : For displacement calculations, between 0 - to 4 , area of the positive trapesium $= \frac{1}{2} \times (2+4) \times 4 = 12$

between 4 - to 8, area of negative trapesium $= \frac{1}{2} \times (2+4) \times (-2) =$ -6.

So , the answer is $+6$

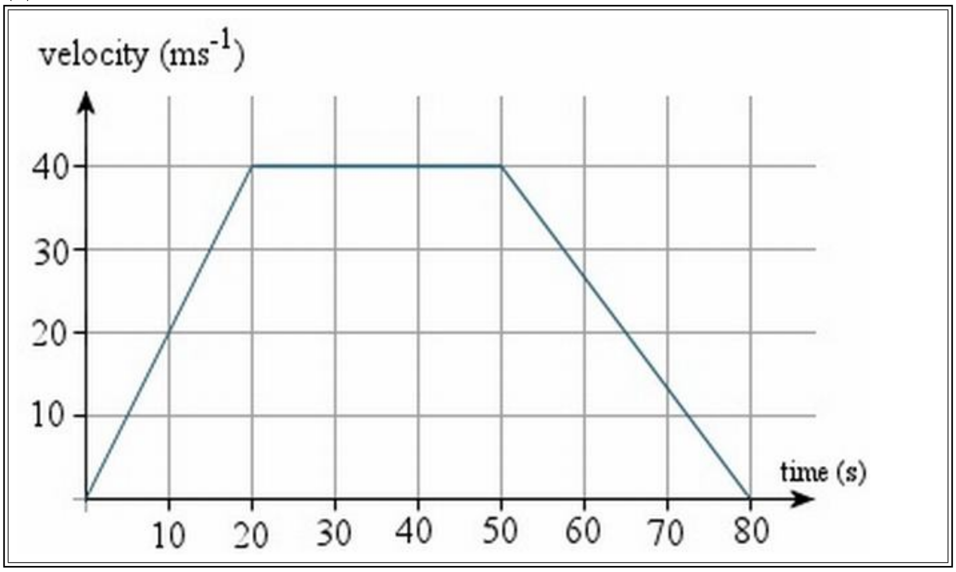
So , a is the correct option.}

Example : A speedboat starts from rest, accelerating at 2 ms -2 for 20 s. It then continues at a steady speed for a further 30 s and decelerates to rest in 30 s. Find:

(a) the distance travelled in m,

(b) the average speed in ms $^{-1}$ and,

(c) the time taken to cover half the distance.



Solution : a) distance=area of trapezium $= \dfrac{(a+b)\,h}{2}$

$= \dfrac{(80+30)\,(40)}{2}$

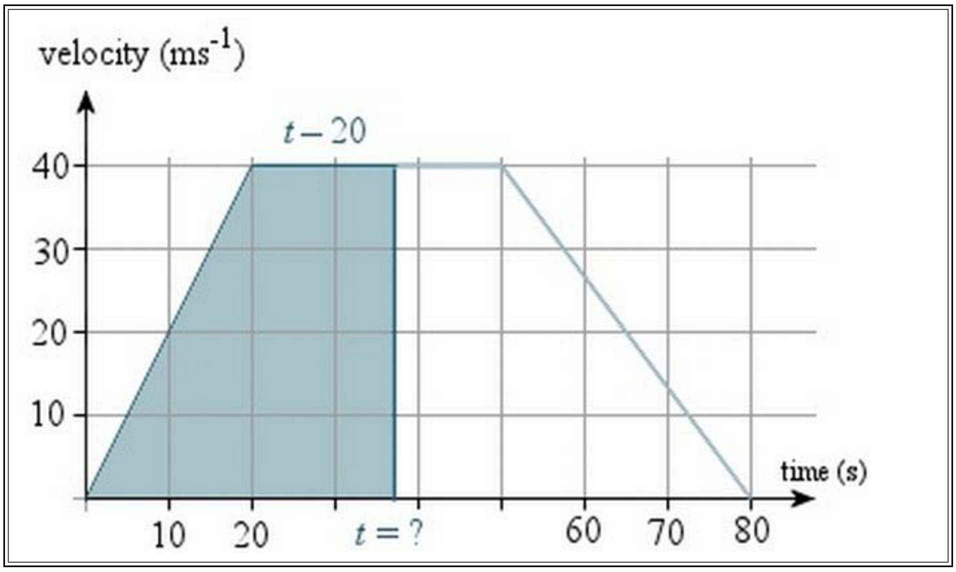
=2200 m

b) average speed $= \dfrac{\text{distance travelled}}{\text{time taken}}$

$= \dfrac{2200}{80}$

=27.5 ms $^{-1}$

c) We need to find the time when the area of the trapezium is half of its original area, or 1100m , as shown in the graph.



The base of this unknown trapezium has length \displaystyle{t}t, and the top of the trapezium will have length t−20. So we have:

area of trapezium $= \dfrac{(a+b)\,h}{2}$

$1100 = \dfrac{(t+[t-20])\,40}{2} = 20\,(2t-20)$

55=2t−20

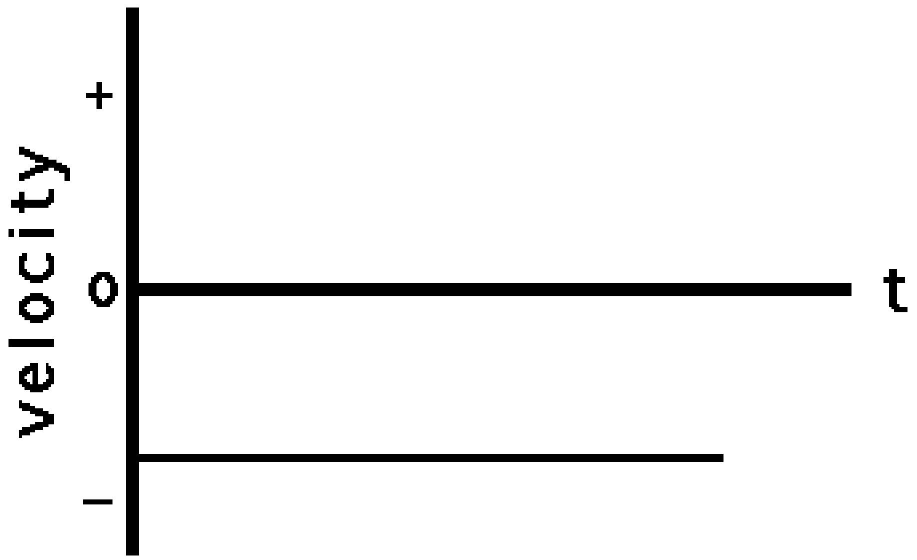
75=2t

t=37.5s

So it will take 37.5 s to cover half the distance.

1. Sketch a velocity-time graph for an object moving with a constant speed in the negative direction.

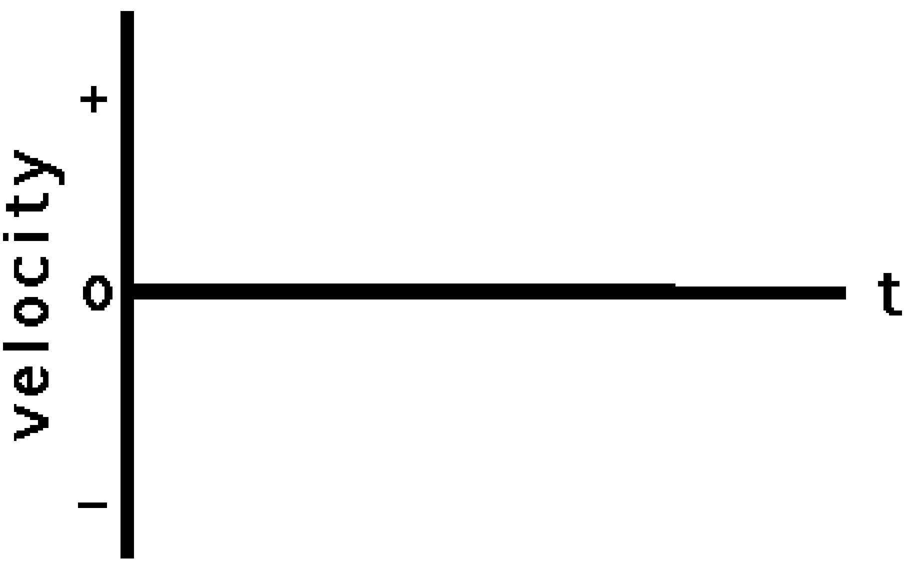
Answer : A velocity-time graph for an object moving with a constant speed in the negative direction is shown below. To have "a constant speed in the negative direction" is to have a - velocity which is unchanging. Thus, the line on the graph will be in the - region of the graph (below 0). Since the velocity is unchanging, the line is horizontal. Since the slope of a line on a v-t graph is the object's acceleration, a horizontal line (zero slope) on a v-t graph is characteristic of a motion with zeo acceleration (constant velocity).



Graph of Question 28

2. Sketch a velocity-time graph for an object which is at rest.

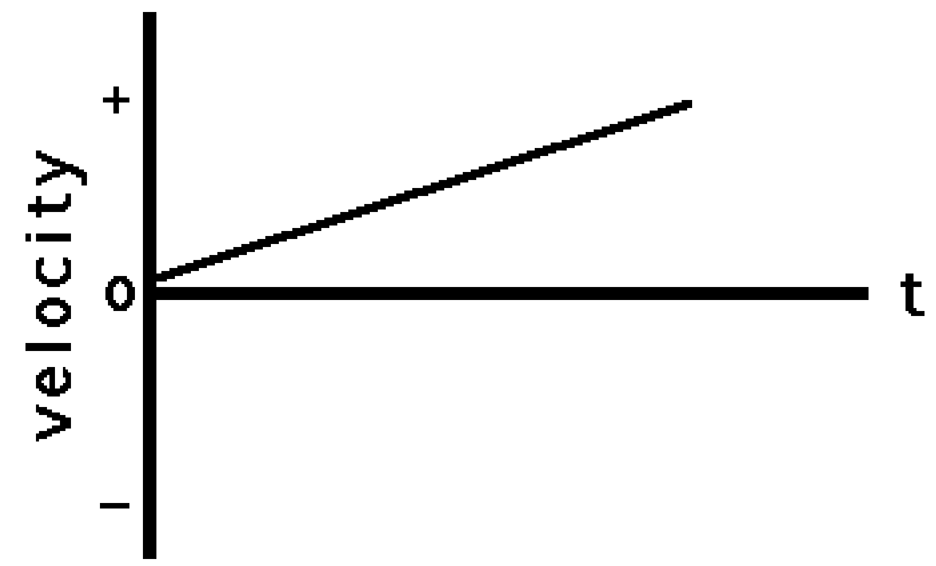
Answer : A velocity-time graph for an object which is at rest is shown below. To be "at rest" is to have a zero velocity. Thus the line is drawn along the axis (v=0).



Graph of Question 29

3. Sketch a velocity-time graph for an object moving in the + direction, accelerating from a slow speed to a fast speed.

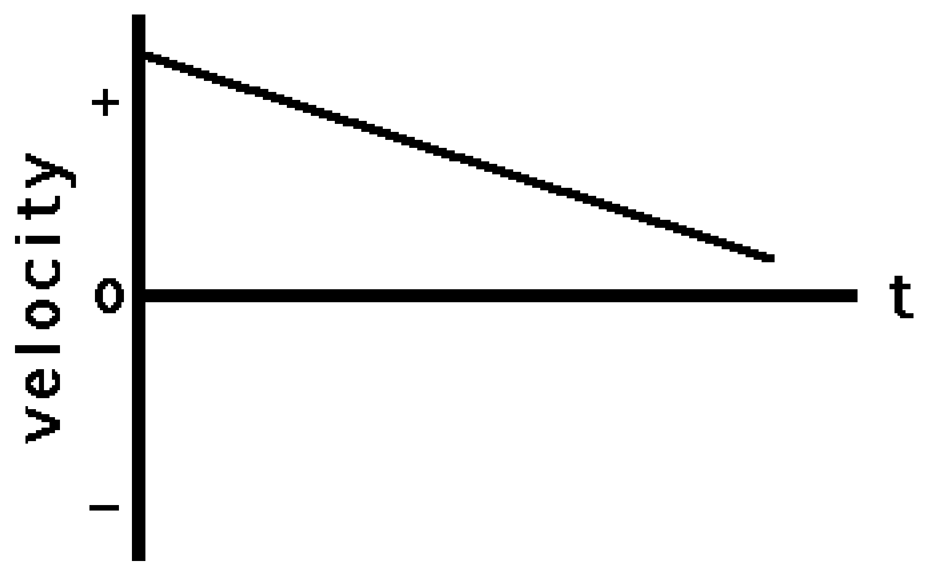
Answer : A velocity-time graph for an object moving in the + direction, accelerating from a slow speed to a fast speed is shown below. An object which is moving in the + direction and speeding up (slow to fast) has a + acceleration. (If necessary, review the dir'n of the acceleration vector in the Physics Classroom Tutorial.) Since the slope of a line on a v-t graph is the object's acceleration, an object with + acceleration is represented by a line with + slope. Thus, the line is a straight diagonal line with upward (+) slope. Since the velocity is +, the line is plotted in the + region of the v-t graph.



Graph of Question 30

4. Sketch a velocity-time graph for an object moving in the + direction, accelerating from a fast speed to a slow speed.

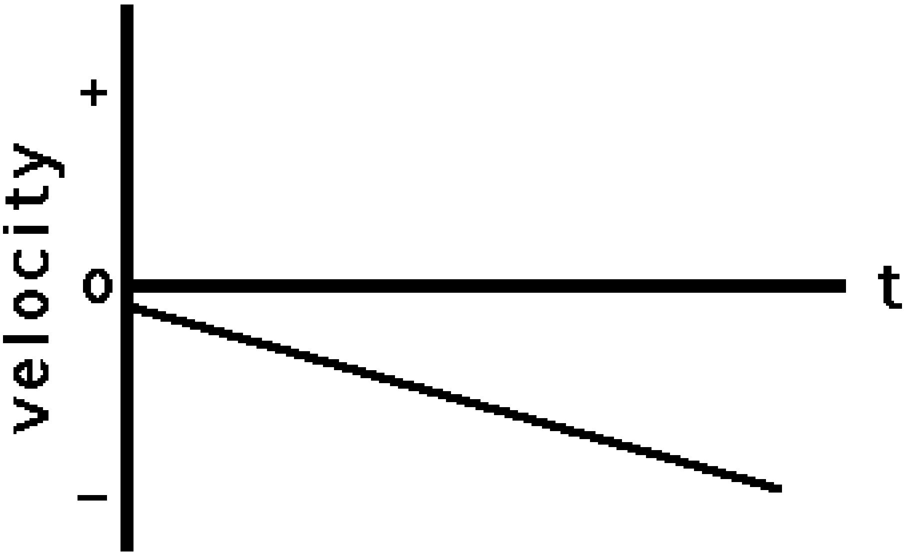
Answer : A velocity-time graph for an object moving in the + direction, accelerating from a fast speed to a slow speed is shown below. An object whgich is moving in the + direction and slowing down (fast to slow) has a - acceleration. (If necessary, review the dir'n of the acceleration vector in the Physics Classroom Tutorial.) Since the slope of a line on a v-t graph is the object's acceleration, an object with - acceleration is represented by a line with - slope. Thus, the line is a straight diagonal line with downward (-) slope. Since the velocity is +, the line is plotted in the + region of the v-t graph.



Graph of Question 31

5. Sketch a velocity-time graph for an object moving in the - direction, accelerating from a slow speed to a fast speed.

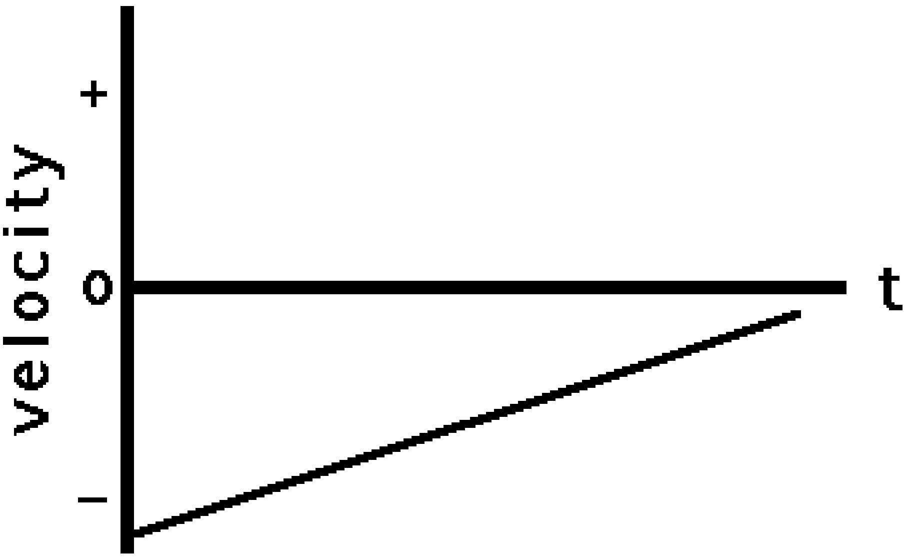
Answer : A velocity-time graph for an object moving in the - direction, accelerating from a slow speed to a fast speed is shown below. An object whgich is moving in the - direction and speeding up (slow to fast) has a - acceleration. (If necessary, review the dir'n of the acceleration vector in the Physics Classroom Tutorial.) Since the slope of a line on a v-t graph is the object's acceleration, an object with - acceleration is represented by a line with - slope. Thus, the line is a straight diagonal line with downward (-) slope. Since the velocity is -, the line is plotted in the - region of the v-t graph.



Graph of Question 32

6. Sketch a velocity-time graph for an object moving in the - direction, accelerating from a fast speed to a slow speed.

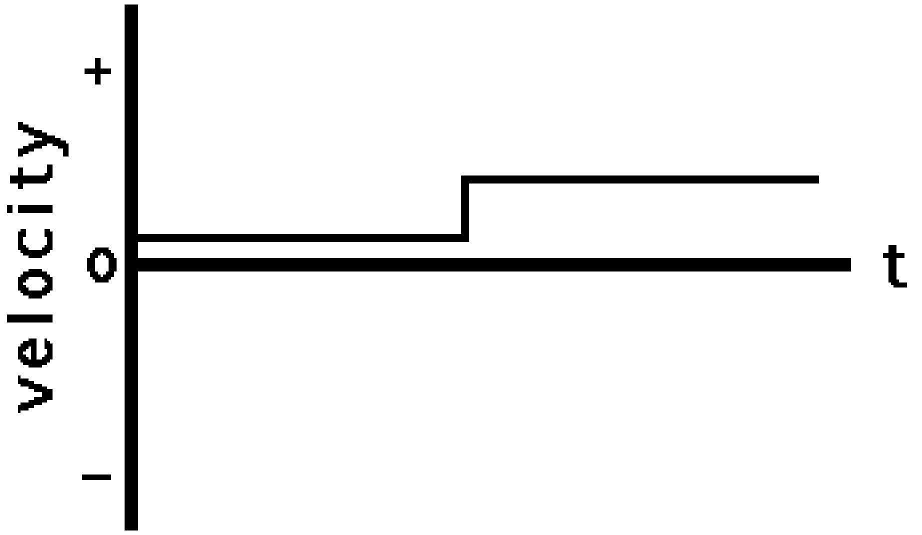
Answer : A velocity-time graph for an object moving in the - direction, accelerating from a fast speed to a slow speed is shown below. An object whgich is moving in the - direction and slowing down (fast to slow) has a + acceleration. (If necessary, review the dir'n of the acceleration vector in the Physics Classroom Tutorial.) Since the slope of a line on a v-t graph is the object's acceleration, an object with + acceleration is represented by a line with + slope. Thus, the line is a straight diagonal line with upward (+) slope. Since the velocity is -, the line is plotted in the - region of the v-t graph.



Graph of Question 33

7. Sketch a velocity-time graph for an object which first moves with a slow, constant speed in the + direction, and then with a fast constant speed in the + direction.

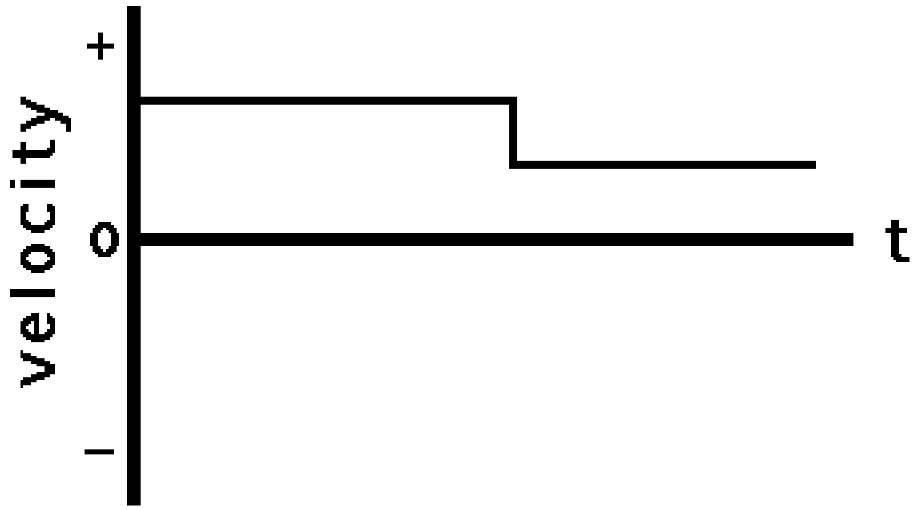
Answer : A velocity-time graph for an object which first moves with a slow, constant speed in the + direction, and then with a fast constant speed in the + direction is shown below. Since there are two parts of this object's motion, there will be two distinct parts on the graph. Each part is in the + region of the v-t graph (above 0) since the velocity is +. Each part is horizontal since the velocity during each part is constant (constant velocity means zero acceleration which means zero slope). The second part of the graph will be higher since the velocity is greater during the second part of the motion.



Graph of Question 34

8. Sketch a velocity-time graph for an object which first moves with a fast, constant speed in the + direction, and then with a slow constant speed in the + direction.

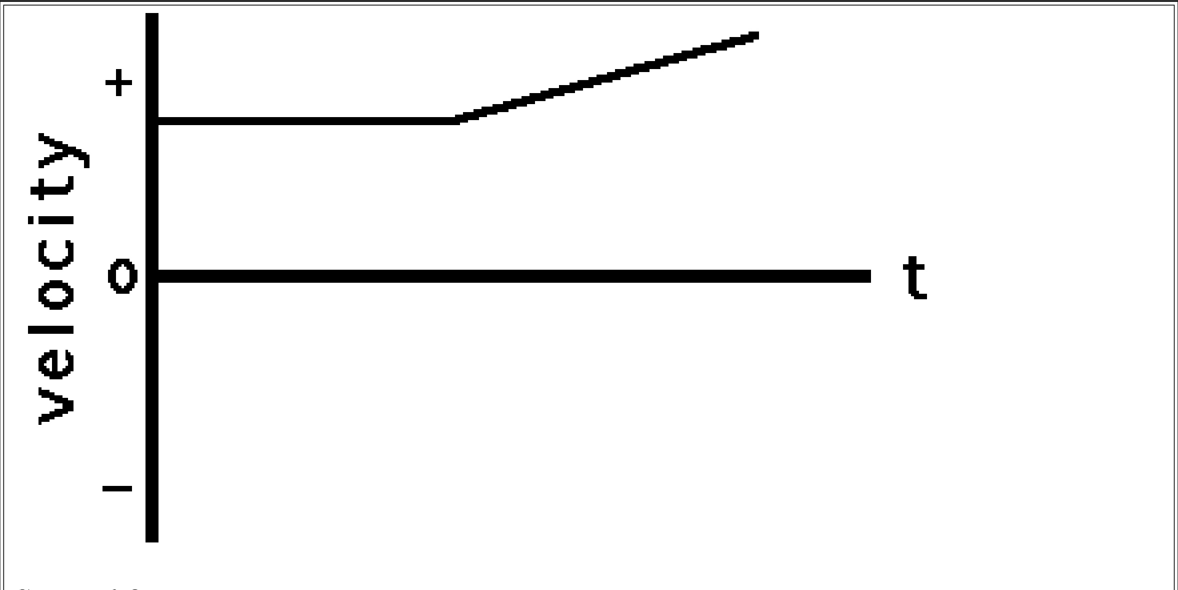
Answer : A velocity-time graph for an object which first moves with a fast, constant speed in the + direction, and then with a slow constant speed in the + direction is shown below. Since there are two parts of this object's motion, there will be two distinct parts on the graph. Each part is in the + region of the v-t graph (above 0) since the velocity is +. Each part is horizontal since the velocity during each part is constant (constant velocity means zero acceleration which means zero slope). The first part of the graph will be higher since the velocity is greater during the first part of the motion.



Graph of Question 35

7. Sketch a velocity-time graph for an object which first moves with a constant speed in the + direction, and then moves with a positive acceleration.

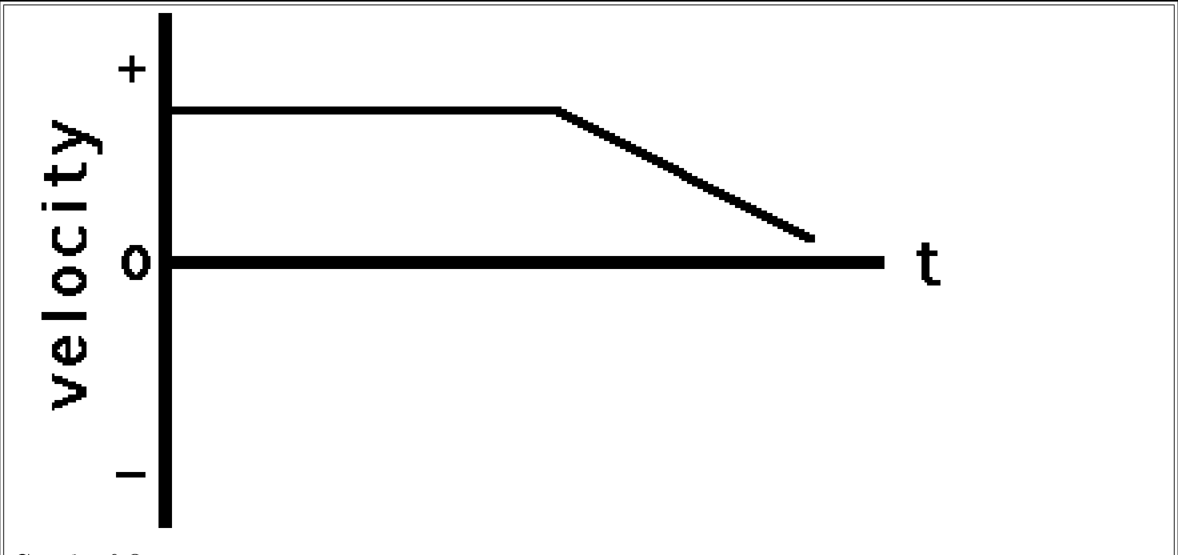
Answer : A velocity-time graph for an object which first moves with a constant speed in the + direction, and then moves with a positive acceleration is shown below. Since there are two parts of this object's motion, there will be two distinct parts on the graph. Each part is in the + region of the v-t graph (above 0) since the velocity is +. The slope of the first part is zero since constant velocity means zero acceleration and zero acceleration is represented by a horizontal line on a v-t graph (slope = acceleration for v-t graphs). The second part of the graph is an upward sloping line since the object has + acceleration (again, the slope = acceleration for v-t graphs)



Graph of Question 36

8. Sketch a velocity-time graph for an object which first moves with a constant speed in the + direction, and then moves with a negative acceleration.

Answer : A velocity-time graph for an object which first moves with a constant speed in the + direction, and then moves with a negative acceleration is shown below. Since there are two parts of this object's motion, there will be two distinct parts on the graph. Each part is in the + region of the v-t graph (above 0) since the velocity is +. The slope of the first part is zero since constant velocity means zero acceleration and zero acceleration is represented by a horizontal line on a v-t graph (slope = acceleration for v-t graphs). The second part of the graph is an downward sloping line since the object has - acceleration (again, the slope = acceleration for v-t graphs)



Graph of Question 37